

Prisha Singh

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EDUCATION

University of Illinois Urbana-Champaign

GPA: 3.94/4.0

Bachelor of Science in Computer Science and Mathematics, Minor in Statistics

August 2023 – May 2027

- **Honors:** Dean's List, James Scholar Honors, Engineering City Scholar, CS Research Scholar & Student Ambassador
- **Coursework:** Applied Machine Learning, Systems Programming, Networks, Advanced Algorithms, Databases

SKILLS

Languages: Python, C++, Java, SQL, R, C, MATLAB, JavaScript/HTML/CSS

Frameworks & Libraries: TensorFlow, Flask, NumPy, pandas, SciPy, scikit-learn

Tools & Platforms: AWS (EC2, IoT, Lambda), GCP, Azure, Docker, Git, MySQL, CI/CD

EXPERIENCE

Oracle Cloud Infrastructure (OCI)

May 2026 – Aug 2026

Software Engineer Intern

Seattle, Washington

- Built a RAG assistant for 70+ OCI engineers, cutting source-discovery time by 87.9% versus ChatGPT with connectors.
- Developed a Node.js MCP server (JSON-RPC) for Shared Knowledge Store API, implementing graph-hop expansion, reranking, and top-*k* retrieval with metadata-preserving citations across 16.7K+ internal repositories and docs.
- Engineered a permission-aware retrieval layer decoupled from generation, achieving 0 errors and 100% corpus isolation.
- Optimized retrieval via A/B pipeline benchmarking, achieving 263 ms p95 latency over 200 requests.

Cognizant × Amazon Web Services (AWS)

Jan 2026 – May 2026

Artificial Intelligence Intern, Manufacturing Innovation Center

Chicago, Illinois

- Replaced a hardcoded defect detector with a CV pipeline (Flask, RealSense, Tulip), benchmarking 6 ML approaches before deploying an AWS assembly copilot on NVIDIA Jetson for real-time edge inference, achieving >85% OEE.
- Engineered a modular CV streaming pipeline (RTSP, Python, FFmpeg, MediaMTX), extending PLC/MQTT infra.
- Built an IoT pipeline streaming 20+ sensor tags into Omniverse digital twins; led Meta Quest VR demos for clients.

Golub Capital

Jun 2025 – Oct 2025

Software Engineer Intern

Chicago, Illinois

- Built a data-protection dashboard across 11 departments (SQL, Python ETL, Power BI) and automated data-governance pipelines in Azure Data Factory, driving a 64% increase in file-labeling compliance.

University of Illinois Urbana-Champaign

Aug 2024 – Present

Course Assistant, Department of Statistics

Urbana, Illinois

- Instructed 200+ students in data analytics, visualization, and machine learning using NumPy, SciPy, and pandas.
- Engineered 3+ full-stack sites (JavaScript/HTML/CSS, MySQL) serving 11,000+ faculty and students annually.

PROJECTS & RESEARCH

FinTech Lab @ UIUC – Lossless PCAP Compression

Jan 2025 – May 2025

- Built a C++ libpcap tool to losslessly compress IEX DEEP PCAP feeds, achieving up to 95% size reduction.
- Designed price/time field-specific compression using delta encoding, header compaction, and LZ4/Zstd entropy coding.
- Designed modular parsers with automated round-trip checks validating compression ratio and correctness.

Illinois Geometry Lab – Spatial Spreading of Altruism

Jan 2025 – May 2025

- Developed agent-based and mean-field PDE simulations (MATLAB) modeling competition, motility, and altruism.
- Proved weak convergence of agent-based models to mean-field limits and analyzed emergent spatial patterns.
- Presented research findings at the Illinois Undergraduate Research Symposium as part of a 4-person team.

TECHNICAL LEADERSHIP

Women in Computer Science, Corporate Committee Board Member

Aug 2023 – Aug 2024

- Secured \$25K+ in corporate partnerships for 100+ female members via outreach to Google, Uber, and IMC Trading.

Disruption Lab, Gies College of Business, Technical Project Manager

Jan 2024 – Dec 2024

- Led a team of 6; designed an agentic AI chatbot for diabetes patients for client OSF Healthcare using AWS Lambda.